

Week 3 – Quantum Computing

Bloch sphere, Measurement, Interference

Vikram Kamat

Review from Week 2

- A qubit is the simplest physical system / unit of information/computing in quantum computing, and is represented as $|\psi\rangle = a|\phi_0\rangle + b|\phi_1\rangle$, where $\{|\phi_0\rangle, |\phi_1\rangle\}$ is an orthonormal basis for $\mathbb{C}^2$, and $|a|^2 + |b|^2 = 1$.
- The standard/computational basis is $\{|0\rangle, |1\rangle\}$, where $|0\rangle = (1 \ 0)^T$ and $|1\rangle = (0 \ 1)^T$.
- Global phase is physically indistinguishable and hence irrelevant, i.e. a qubit $|\psi\rangle$ is physically indistinguishable from $e^{i\phi}|\psi\rangle$, where $e^{i\phi}$ is any complex number of unit modulus.
- **Unitary evolution:** Qubits can be transformed by unitary operators/matrices/gates. Unitary matrices are norm-preserving and inner product preserving.
- Some examples of single-qubit unitary gates: $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, $Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, $H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$. X, Y, Z are called as the Pauli gates, and H is called as the Hadamard Gate. We will see other one-qubit (as well as two-qubit, and higher qubit gates) soon.

The Bloch Sphere

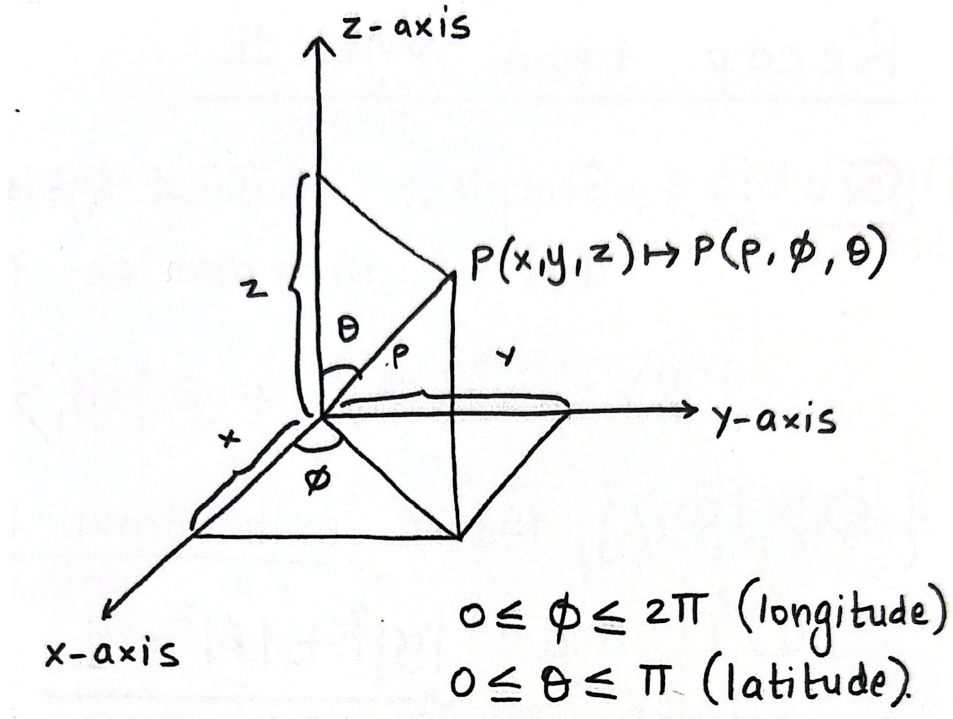
The Bloch sphere presents a useful visual abstraction for understanding single qubit states, as well as the action of unitary transformations on single qubit states. Broadly, we will plot any qubit $|\psi\rangle = a|0\rangle + b|1\rangle$ on the unit sphere. See Figure 1 below to observe the spherical coordinates (ρ, ϕ, θ) of a random point on the unit sphere. Consider $|\psi\rangle = a|0\rangle + b|1\rangle$, with $|a|^2 + |b|^2 = 1$. Writing the complex numbers $a = |a|e^{i\phi_0}$ and $b = |b|e^{i\phi_1}$ in polar form, factoring out an (indistinguishable) global phase of $e^{-i\phi_0}$, and rewriting $\phi = \phi_1 - \phi_0$ we obtain $|\psi\rangle = |a||0\rangle + |b|e^{i\phi}|1\rangle$. Finally, since $|a|^2 + |b|^2 = 1$, we can write $|a| = \cos(\theta/2)$, and $|b| = \sin(\theta/2)$ to obtain the final form

$$|\psi\rangle = \cos(\theta/2)|0\rangle + \sin(\theta/2)e^{i\phi}|1\rangle.$$

This now corresponds to a vector $(\rho = 1, \phi, \theta)$ on the unit sphere, as shown in Figure 2. Note that the latitudinal angle span is effectively halved, as $0 \leq \theta \leq \pi$ implies that $0 \leq \theta/2 \leq \pi/2$. The following exercise provides more insight into this matter.

Exercise 1. In the Bloch sphere representation, a pure qubit state is written as

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle, \quad 0 \leq \theta \leq \pi, \quad 0 \leq \phi < 2\pi.$$



Scanned with
CamScanner

Figure 1: Spherical Coordinates in 3-D.

Using the physical indistinguishability of global phase, explain why values of θ outside the interval $[0, \pi]$ are not needed.

Hint: For $\theta' = 2\pi - \theta$, with $\theta' < \pi$ for $\theta > \pi$, find ϕ' such that the qubit with Bloch sphere representation corresponding to angles ϕ' and θ' is equivalent (up to global phase) to $|\psi\rangle$.

The following exercise asks you to find Bloch sphere representations (in terms of ϕ and θ values) for each of the single qubit states.

Exercise 2. Find $|\phi\rangle$ and $|\theta\rangle$ values corresponding to the Bloch sphere representations for:

$$1. |+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

$$2. |-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$$

$$3. |\psi_1\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle$$

$$4. |\psi_2\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{i}{2}|1\rangle$$

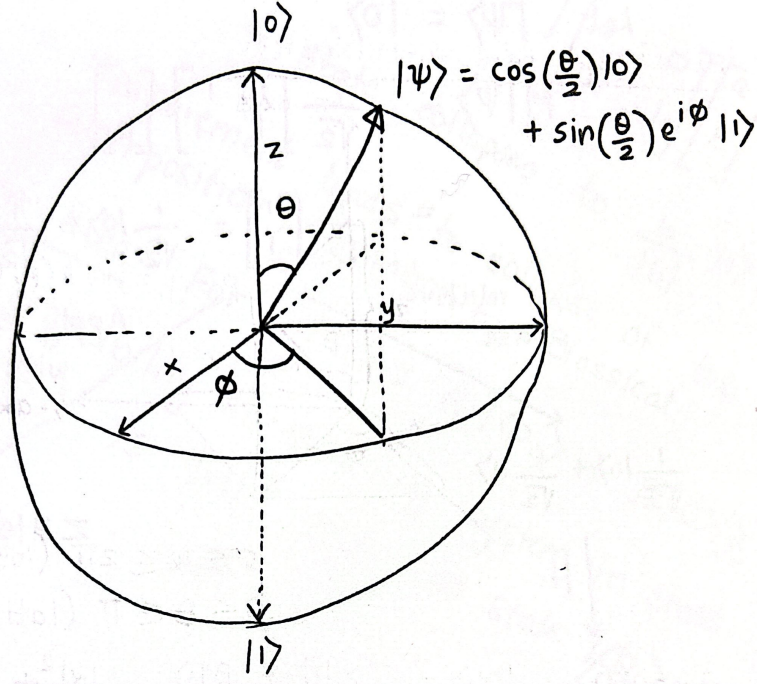


Figure 2: Pure Qubit on the Bloch Sphere.

Measurement

Measuring a quantum state is done by a collection of measurement operators $\{M_m\}$ – for single-qubit states, $m \in \{0, 1\}$. Given a single qubit state $|\psi\rangle = a|\phi_0\rangle + b|\phi_1\rangle$, where $\{|\phi_0\rangle, |\phi_1\rangle\}$ is any orthonormal basis for $\mathbb{C}_2$, the probability of measuring outcome $|\phi_m\rangle$ $m \in \{0, 1\}$ is given by

$$p(m) = \langle\psi|M_m^\dagger M_m|\psi\rangle.$$

The measurement operators also have to obey the following **completeness** property:

$$\sum_m M_m^\dagger M_m = I.$$

From probability theory, we must also have

$$\sum_m p(m) = \sum_m \langle\psi|M_m^\dagger M_m|\psi\rangle = 1.$$

Finally, measurement leads to a **collapse** of the superposition, and the **post-measurement** state is given by

$$\frac{M_m|\psi\rangle}{\sqrt{\langle\psi|M_m^\dagger M_m|\psi\rangle}}$$

The measurement mechanism becomes more transparent when we measure in the computational basis. The two **projective** measurement operators are $M_0 = |0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $M_1 = |1\rangle\langle 1| =$

$\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$. Note that the projective measurement operators for any orthonormal basis $\{\phi_0, \phi_1\}$ are computed using the outer products $|\phi_i\rangle\langle\phi_i|$, $i \in \{0, 1\}$. Indeed, these also naturally extend to multi-qubit measurements, as we will see later.

Born rule. For a given qubit $|\psi\rangle = a|0\rangle + b|1\rangle$, with $|a|^2 + |b|^2 = 1$, the probability of measuring $|0\rangle$ (alternately, $|1\rangle$) is $|a|^2$ (alternately, $|b|^2$). Furthermore, if the outcome observed is, say, $|0\rangle$, then the post-measurement system is $\frac{a}{|a|}|0\rangle$ (which is, of course, equivalent to $|0\rangle$). Verify that these calculations agree with the general definitions of measurement probabilities and post-measurement states stated earlier.

Exercise 3.

1. For a single qubit state $|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle$, compute the respective probabilities of measurement $|0\rangle$ and $|1\rangle$.
2. Compute the measurement operators M_+ and M_- for the orthonormal basis $\{|+\rangle, |-\rangle\}$.

A brief introduction to interference

To end this lecture, we will circle back to our brief encounter with the (in)famous double slit experiment. While a complete understanding of its nuances, intricacies and implications are likely outside the scope of these notes, we will explain it using the lens of quantum interference. For that, let us examine two scenarios, where a single-qubit system is acted upon by a sequence of quantum gates. In both instances, suppose you begin with $|0\rangle$.

1. In the first instance, suppose you apply the Hadamard gate **twice** to $|0\rangle$. Since it is easily verified that $H^2 = I$, the intermediate stage of the qubit will be $|+\rangle$, but the final state of the qubit will again be $|0\rangle$.
2. In the second instance, suppose you apply the Hadamard gate, the the Pauli Z gate, and the Hadamard gate again. $H|0\rangle = |+\rangle$, $Z(H|0\rangle) = Z|+\rangle = |-\rangle$, and finally $H(Z(H|0\rangle)) = H|-\rangle = |1\rangle$.

The former scenario is an example of constructive interference, while the latter is an example of destructive interference.

With regards to the double slit experiment, we merely wish to highlight the shared logical structure with the quantum circuit example above. In the double slit experiment, photons are fired one at a time toward a screen with two slits, and each photon is detected at a single position on a distant screen. When both slits are open, the resulting distribution of detection locations exhibits regions where the probability of detection is lower than when only a single slit is open, a phenomenon incompatible with classical probability theory.

This mirrors the mechanism underlying quantum interference in our examples above. When no measurement is performed to determine which slit a photon passes through, the alternatives corresponding to the two slits are not assigned probabilities independently. Instead, each alternative contributes a complex probability amplitude, and these amplitudes are summed before measuring

observable probabilities. Depending on their relative phases, this sum can result in either constructive or destructive interference.

This is mathematically analogous to applying a Hadamard gate, allowing the system to evolve without measurement, and then recombining the resulting amplitudes with a second Hadamard gate. In both cases, intermediate alternatives exist in the quantum description, but are not distinguished by measurement. As a result, interference effects arise that have no classical probabilistic explanation.